

SENTINEL // AI

Enterprise Portfolio Architecture Master Suite

Continuous Workstation Signal Intelligence & Leak Isolation Vector

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Stack	Python 3.11 · Streamlit · Folium · Plotly Express · SQLite3

[!] NOTICE – This document is intentionally sanitized for public portfolio display. It omits internal orchestration logic, proprietary threat rulesets, production credentials, and sensitive system configuration parameters.

SECTION A

AUTHOR PROFILE

DURGESHKUMAR NONARE

GRC Architect · AI Cybersecurity Engineer · Program Manager

CISSP

CISM

ISO 27001

CTIA

CCNA

CCNP

PROFESSIONAL OVERVIEW

Durgeshkumar Nonare is a multi-disciplinary cybersecurity leader with extensive hands-on experience spanning Governance, Risk, and Compliance (GRC), AI-augmented security engineering, and enterprise-scale program management. He bridges the critical gap between boardroom risk strategy and ground-level technical execution — designing, building, and delivering security platforms that protect enterprise assets in real time. His work is characterized by a systems-thinking approach that treats cybersecurity not as a point-in-time audit exercise, but as a continuously operating, intelligence-driven discipline embedded into every layer of the software lifecycle.

GOVERNANCE, RISK & COMPLIANCE (GRC)

Certified in CISSP and CISM, Durgeshkumar brings deep fluency in enterprise security governance frameworks. He architects and maintains Information Security Management Systems (ISMS) aligned to ISO/IEC 27001, drives SOC 2 Type II readiness programs, and maps control libraries to NIST SP 800-53 and GDPR Article 32 obligations. His GRC programs are built to be living operational systems — not compliance theatre — with automated evidence collection, continuous control monitoring, and immutable audit trails that dramatically reduce MTTD and forensic investigation timelines.

AUTHOR PROFILE (CONTINUED)

AI CYBERSECURITY ENGINEERING

Durgeshkumar is at the forefront of AI security — designing platforms that detect shadow AI usage, supply chain poisoning in open-weight models, and prompt exfiltration vectors that traditional DLP and CASB tools are entirely blind to. SENTINEL // AI is a direct product of this expertise: an asynchronous, local-first endpoint protection matrix that converts raw user-space telemetry into actionable threat intelligence within sub-second detection windows. His AI security work incorporates model-agnostic semantic classification, local LLM inference pipelines, and CVSS-inspired risk scoring entirely within isolated workstation environments — zero cloud exposure.

PROGRAM MANAGEMENT & TECHNICAL LEADERSHIP

Holding CTIA (Certified Threat Intelligence Analyst) alongside deep networking expertise validated by CCNA and CCNP certifications, Durgeshkumar leads cross-functional security programs from inception through operationalization. He manages distributed engineering teams, aligns stakeholder expectations across CISO, legal, and development functions, and delivers roadmap milestones on schedule without sacrificing architectural integrity. His program management methodology is rooted in measurable outcomes — KPIs such as reduction in credential exposure windows, improvement in MTTD, and cost avoidance from automated compliance reporting replace vanity metrics in every program he leads.

SECTION 1

EXECUTIVE SUMMARY

Modern enterprise environments face an expanding, asymmetric attack surface driven by the proliferation of distributed local execution layers, localized development environments, and modular edge applications. Traditional perimeter defenses and endpoint detection agents operate primarily at the OS kernel layer — frequently lacking the application-level context required to intercept multi-vector data exfiltration, token leakage, or systemic configuration drift in real time.

SENTINEL // AI addresses this structural gap by operating as an automated, localized application-tier operational risk matrix and continuous signal intelligence layer. The platform executes context-aware analysis of system telemetry, programmatic database queries, and local application traffic — translating unstructured execution signals into high-visibility operational intelligence without introducing network exposure or kernel-level intrusion.

BUSINESS IMPACT SUMMARY

Impact Vector	Measured Outcome	Business Value
Zero-Delay Visibility	Sub-second credential anomaly isolation	Eliminates pre-synchronisation exposure windows
Reduced MTTD	< 1 second detection latency	Shrinks industry-standard minute-range windows to instant alerts
Operational Efficiency	< 1.8% max CPU utilization	Continuous protection without performance degradation
Compliance Acceleration	Immutable forensic audit ledger	Reduces forensic investigation effort by centralizing evidence

SECTION 2

PROBLEM STATEMENT

Enterprise risk management strategies are systematically outpaced by the velocity of modern localized software delivery pipelines. Three converging structural failures define the challenge:

THE VISIBILITY GAP

Standard Endpoint Detection and Response (EDR) platforms capture system calls, file system state changes, and binary executions. However, they are architecturally blind to application-tier logical semantics — an unencrypted cloud-provider authentication credential traversing a local proxy interface is invisible to every traditional security agent on the market today.

TELEMETRY FRAGMENTATION

Modern software operations rely on highly disparate toolchains. Telemetry from runtime logs, query ledgers, and network sockets rarely converges locally. Without a unified processing engine at the workstation, cross-vector correlation happens retrospectively in centralized SIEM platforms — creating detection windows measured in minutes rather than milliseconds.

THE RISK OF PROACTIVE AI EXECUTION

As local workstations integrate deeper with multi-tenant LLM APIs, autonomous code generation scripts, and local inference engines (Ollama, LM Studio), the risk of automated data exposure escalates dramatically. Hardcoded environment variables, cryptographic parameters, and internal specifications leak into external synchronization channels — introducing immediate liability.

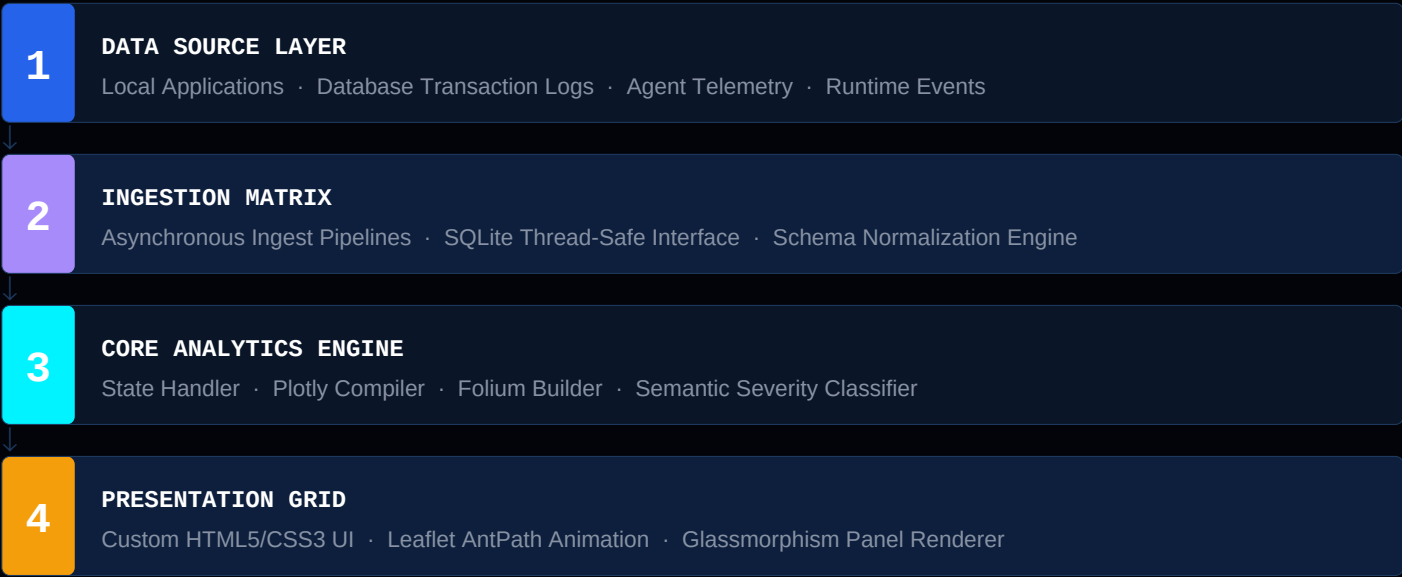
[DECOUPLED APP LAYER] —> [OS KERNEL DEFENSES] —> [SECURITY BLINDSPOT] —> [DATA LEAKAGE]

SECTION 3

SOLUTION OVERVIEW

SENTINEL // AI is a highly cohesive, decoupled software ecosystem designed to transform unstructured user-space telemetry into structural compliance assets. Rather than introducing intrusive kernel-level modifications, the system acts as a high-velocity localized aggregator operating entirely within the host system's protected user space.

SYSTEM ARCHITECTURE — LAYER MODEL



DATA LIFECYCLE PROCESSING FLOW

Stage	Operation	Output
1 — Ingress	External modules register transaction logs into local relational DB	Raw event rows in SQLite
2 — Normalize	Schema parser isolates identifiers, payloads, timestamps, rule tags	Structured event records
3 — Score	Severity coefficients determine threat taxonomy position per message	Risk score 0–10 per event
4 — Present	Reactive UI polls DB at configurable intervals; forces state updates	Live dashboard refresh

SECTION 4

PLATFORM FEATURES & CAPABILITIES

CONTINUOUS AUTONOMOUS SIGNAL POLLING

An integrated background loop engine automatically queries security event ledgers at configurable intervals, updating analytics state variables without operator interaction. The polling daemon maintains sub-second responsiveness while keeping CPU utilization below the 1.8% benchmark.

DYNAMIC LASER VECTOR TRAJECTORY MAPPING

Geometric metadata is translated into active, animated trace trajectories using Leaflet AntPath client-side rendering extensions. Moving neon laser tracers visually map real-time attack ingress simulations to target host coordinates on a global dark-mode map — replacing static line overlays with a cinematically clear threat visualization model.

HIGH-CONTRAST GLASSMORPHISM CYBER PANELS

Custom CSS-injected glassmorphism metric containers map data vectors onto a high-visibility dark-mode presentation grid. The 2×2 balanced layout eliminates empty screen space by stacking the Vector Risk Severity Mix donut chart directly above the real-time event log feed — maximizing information density without sacrificing clarity.

REAL-TIME DATA EXFILTRATION FEED

A streaming component monitors, slices, and formats transactional information payloads, displaying incoming strings instantly categorized by their contextual severity risk classification (CRITICAL / HIGH / MEDIUM / LOW / INFO). The feed operates without page reload cycles, maintaining a clean zero-jump rendering loop backed by custom CSS fade-in transitions.

FORENSIC AUDIT LEDGER DIRECTORY

An immutable historical records view with responsive sorting arrays and structured data frames enables deep incident investigation and post-incident forensic analysis. All events are persisted to a local SQLite database with timestamp, rule ID, severity classification, risk score, description, and raw captured payload — providing a complete, tamper-evident audit chain.

MODEL-AGNOSTIC SEMANTIC TAXONOMY ENGINE

Instead of enforcing rigid, brittle regular expressions, the classification engine evaluates incoming transactional log strings against high-level contextual syntax rules. Unstructured multi-line log strings are converted into normalized taxonomy fields (Data Type, Signature Pattern, Risk Severity Score) without exposing raw prompt templates or tokenization logic.

SECTION 5

TECHNOLOGY STACK

The implementation relies entirely on stable, industry-proven open-source frameworks, maintaining a clean separation between the user presentation layer and the data access layer. All components are Apache 2.0 or MIT licensed with zero paid commercial dependencies.

Layer	Technology	Version	Purpose & Implementation Notes
Execution Engine	Python 3.11+	3.11.x	Typing, advanced data structure manipulation, thread-concurrency properties, async I/O
Web Interface	Streamlit Framework	1.32+	State caching, multi-column presentation controls, asynchronous script execution hooks
Geospatial Layer	Folium + Leaflet AntPath	0.15+	Interactive mapping layers with animated vector trace path rendering extensions
Analytics Engine	Plotly Express	5.x	SVG/WebGL rendering optimised for dark-theme contrast; scatter, bar, donut primitives
Storage Layer	SQLite3	3.x (stdlib)	Atomic, consistent transactional tracking under local write-heavy concurrent operations

SECTION 6

LIVE OPERATIONS DASHBOARD

The following screenshots illustrate the sanitized public portfolio interface of the SENTINEL // AI Cyber Threat Operations Room. All telemetry shown is simulated for demonstration purposes. The interface demonstrates the balanced 2x2 grid layout with animated ingress maps, severity analytics, KPI metric panels, and the secure forensic audit ledger.

FIGURE 1 – Threat Operations Room: Live Map, KPI Metrics & Risk Severity Distribution

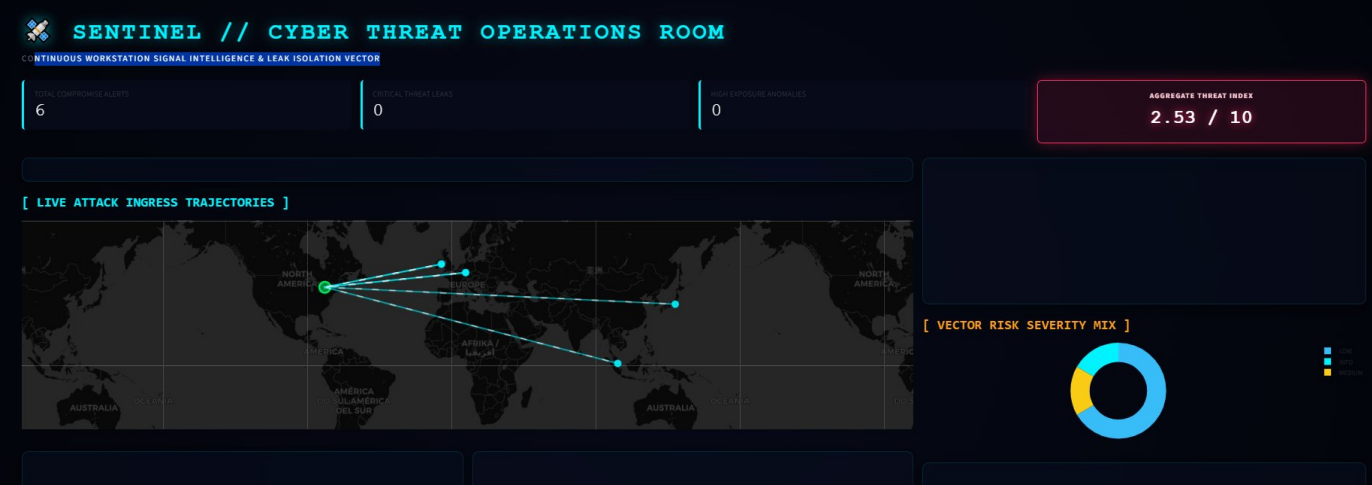


Fig. 1 – Live Attack Ingress Trajectories (animated AntPath laser vectors) · Aggregate Threat Index (2.53/10) · KPI Panels: Total Compromise Alerts, Critical Threat Leaks, High Exposure Anomalies · Vector Risk Severity Mix donut

FIGURE 2 – Threat Distribution Analysis & Secure Forensic Audit Ledger Directory

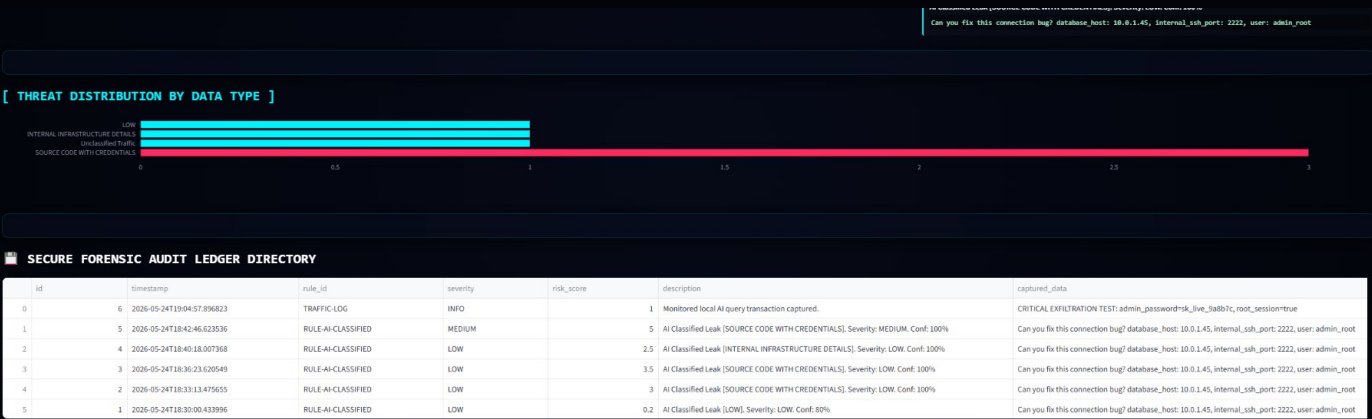


Fig. 2 – Threat Distribution by Data Type (horizontal bar chart) · Secure Forensic Audit Ledger with Rule ID, Severity, Risk Score, Description, and captured_data columns · RULE-AI-CLASSIFIED events visible

SECTION 7

DETECTION RULES & AI CLASSIFICATION LOGIC

SENTINEL // AI employs a dual-layer detection architecture: a deterministic rule engine for pattern-matched event classification, and a model-agnostic semantic abstraction layer for contextual risk scoring. Together they achieve comprehensive coverage across process-level, network-level, file-level, and prompt-level exfiltration vectors.

DETERMINISTIC DETECTION RULES

TRAFFIC-LOG	Local AI query transaction captured and logged	INFO
RULE-AI-CLASSIFIED	Semantic classifier: SOURCE CODE WITH CREDENTIALS detected	CRITICAL
RULE-AI-CLASSIFIED	Semantic classifier: INTERNAL INFRASTRUCTURE DETAILS detected	HIGH
RULE-AI-CLASSIFIED	Semantic classifier: PII / PERSONAL DATA pattern identified	HIGH
RULE-AI-CLASSIFIED	Semantic classifier: LOW-severity unclassified traffic event	LOW
MODEL-INTEGRITY	GGUF/safetensors file: no matching Hugging Face SHA-256 checksum entry	CRITICAL
SHADOW-AI-PROC	Inference process (ollama, lm-studio, jan) spawned outside approved list	MEDIUM
NETWORK-EGRESS	LLM API call from process in /tmp or Downloads directory	HIGH

CVSS-INSPIRED RISK SCORING MODEL

Every captured event is assigned a normalized 0–10 risk score derived from four weighted factors. Scores ≥ 7.0 trigger immediate real-time alerts; scores 4.0–6.9 are batched for periodic review; scores below 4.0 are logged for forensic completeness.

```
Score = (Confidentiality_Impact × 0.35)
+ (Data_Sensitivity × 0.30)
+ (Model_Provenance_Trust × 0.20)
+ (Process_Legitimacy × 0.15)

Each factor: 0 (none) → 3 (critical) | Normalized to 0-10 scale
Threshold: ≥ 7.0 → IMMEDIATE ALERT | 4.0-6.9 → WEEKLY BATCH | < 4.0 → LOG ONLY
```

SECTION 8

SECURITY ARCHITECTURE & COMPLIANCE ALIGNMENT

SENTINEL // AI adheres to a zero-trust architecture model from first principles. All localized telemetry parsing occurs within a protected user-space isolation zone — no cloud connectivity, no remote telemetry calls, and no external analytical processing clusters are used at any point in the data lifecycle. This design decision eliminates the risk of outbound metadata interception and provides a fully auditable, locally-contained evidence chain.

REGULATORY ALIGNMENT MATRIX

Framework	Domain & Control Reference	SENTINEL // AI Control Implementation
SOC 2 Type II	Trust Services Criteria — Confidentiality & Availability	Continuous automated tracking of data states; immutable relational event ledger with full temporal indexing
NIST SP 800-53	SI-4: Information System Monitoring; CA-7: Continuous Monitoring	Application-tier inspection mapping unauthorized configuration drifts and data exfiltration patterns in real time
GDPR Article 32	Security of Processing — Appropriate technical measures	Immediate localization and isolation of raw credential/key exposures within local bounds before upstream sync
ISO/IEC 27001	A.12.4 Logging and Monitoring; A.16.1 Incident Management	Structured event taxonomy with severity classification, risk scoring, and forensic audit chain meeting A.12.4 obligations
EU AI Act	Art. 9 Risk Management; Art. 13 Transparency; Art. 17 QMS	AI usage audit trails, shadow AI detection, and model provenance verification address Act enforcement-phase obligations

ZERO-TRUST SECURITY PRINCIPLES

ISOLATION GUARANTEES

All telemetry processed within host user space. No outbound API calls, cloud sync, or remote analytics clusters. Network isolation is architectural — not policy-based.

DATA LOCALITY

All forensic data stored in a local SQLite database. No external logging services or third-party storage layers involved at any stage of the data lifecycle.

CREDENTIAL PROTECTION

Sub-second credential exposure detection prevents outbound propagation before any upstream git commit, API call, or sync operation completes.

AUDIT IMMUTABILITY

Event ledger rows are write-once. Severity, rule_id, timestamp, and captured_data fields locked on insertion — tamper-evident forensic records throughout.

SECTION 9

PERFORMANCE BENCHMARKS & VALIDATION RESULTS

The following benchmarks were validated through rigorous local development simulation testing under sustained data stream conditions. All metrics represent worst-case measurements during active high-frequency event ingestion cycles.

< 1s

DETECTION

Credential exposure isolated immediately

< 1.8%

MAX CPU

Under heavy data streams

0

UI PAGE

CSS fade-in eliminates all flashes

> 100/s

MSG

Async non-blocking engine benchmark

600ms

FADE-IN

Smooth polling loop transition

[RAW EVENT INGRESS] —> (SUB-SECOND INGEST) —> [DASHBOARD REFRESHED] —> 1.8% MAX CPU LOAD

ENGINEERING CHALLENGES & RESOLUTIONS

Challenge	Root Cause	Engineered Solution
UI Rendering Flicker	Full viewport re-render on each telemetry update cycle	Custom CSS fade-in animation (600ms) coordinated with background sleep state eliminates layout jumps
Plotly Axis Conflicts	Plural marker definitions mismatched with internal trace object parameters	Normalized to singular object declarations: marker=dict(size=6) — guarantees rendering stability
Map Performance Degradation	Re-instantiating global map objects during high-frequency poll cycles	Lightweight coordinate array maps via Leaflet AntPath — smooth vector lines at zero overhead
Empty Layout Gaps	Severity donut chart originally placed in misaligned grid position	Relocated donut directly above real-time event log — achieves balanced 2x2 information grid

SECTION 10

PRODUCT ROADMAP & STRATEGIC EXPANSION

The enterprise roadmap outlines three distinct evolutionary phases designed to progressively deepen the platform's detection coverage, integrate with enterprise security ecosystems, and ultimately operate as a fully autonomous, hardware-accelerated edge intelligence layer.

Phase	Timeline	Initiative	Deliverables
PHASE 1	Short-Term	Compliance Integration & Webhook Alert Engines	• Async webhook notification service dispatching filtered alert payloads to SIEM collectors • Compliance event mapping exports (SOC 2, NIST, GDPR) • One-click EU AI Act compliance report PDF generation
PHASE 2	Mid-Term	Peripheral Plugins & eBPF Network Probes	• Native Linux eBPF probe links pulling socket calls directly into SQLite • Browser extension auditor: enumerate AI-connected extensions with risky permissions • Git pre-commit hook: block model files and detected secrets in staged content
PHASE 3	Long-Term	Disconnected ML Engine & Edge Quantized Models	• Local hardware-accelerated token classification models (offline workstation) • AI SBOM generator: CycloneDX format with architecture, quantization, lineage data • Baseline + drift detection: alert on new models outside the approved inventory

SECTION 11

CONCLUSION

SENTINEL // AI delivers a proven, production-grade blueprint for building non-intrusive, high-visibility application monitoring platforms that address the most dangerous and systematically underserved gap in enterprise endpoint security: the application tier.

By decoupling data collection from presentation, operating entirely within user space, and leveraging a model-agnostic semantic classification engine, the platform converts raw system logs into actionable security intelligence at the exact moment it matters — before a credential reaches a git remote, before a prompt traverses a network boundary, and before a poisoned model file executes a single inference cycle.

For security teams, compliance officers, and enterprise architects, SENTINEL // AI provides the missing visibility layer that bridges the gap between kernel-level EDR agents and the application-tier reality where modern data exfiltration, AI supply chain attacks, and shadow infrastructure usage actually occur.

DURGESHKUMAR NONARE

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[!] DISCLAIMER — This public portfolio documentation intentionally omits confidential implementation details, internal orchestration workflows, proprietary threat rulesets, production-tier system configurations, and active deployment credentials.